



USER MANUAL

VibeCheck

How to measure a plugin, read what the numbers mean, and compare two plugins fairly.

For VibeCheck 0.6 on macOS and Windows · Make Believe Studios

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1 What VibeCheck is

VibeCheck loads an audio plugin outside your DAW, plays test signals through it, and tells you what came back.

It answers four kinds of question:

QUESTION	PAGE
What does this plugin do to the sound?	Graph Analyzer
Is it safe to rely on in a session?	Plugin Health
How do these two plugins differ?	A/B Compare
How much CPU will it cost me?	Performance Profiler

WHAT IT IS NOT

VibeCheck measures. It does not tell you whether a plugin sounds good, and a clean measurement is not the same as a useful tool. A saturator is supposed to distort; a vintage EQ is supposed to bend phase. Use the numbers to understand a plugin, then use your ears to decide.

2 Getting started

Install

SYSTEM	WHAT TO DO
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Mac	Open the disk image and drag VibeCheck into Applications . Needs macOS 11 or later; works on Apple silicon and Intel.
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Windows	Run VibeCheck-Windows-Setup.exe . Needs Windows 10 or 11, 64-bit. A portable zip is also available if you would rather not install.
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The first time you open it

The installers are not signed with a paid certificate, so your computer asks once before running them. Nothing is wrong with the file.

- **Mac:** right-click (or Control-click) VibeCheck in Applications and choose **Open**, then **Open** again. If that option is missing, go to **System Settings → Privacy & Security**, scroll down and click **Open Anyway**.
- **Windows:** when SmartScreen says "Windows protected your PC", click **More info**, then **Run anyway**.

Let it find your plugins

On first launch VibeCheck scans the standard plugin folders. The pill at the top right shows progress and ends on **Library up to date**. Scanning happens in a separate process, so a plugin that crashes while being scanned cannot take VibeCheck down; it is listed as skipped instead.

SYSTEM	FORMATS	WHERE IT LOOKS
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Mac	AudioUnit, VST3	/Library/Audio/Plug-Ins and the same folder in your home Library
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Windows	VST3	C:\Program Files\Common Files\VST3
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COPY-PROTECTED PLUGINS

Plugins that need a licence dongle, an activation or special hardware may refuse to load outside a DAW, or load but pass no audio. VibeCheck reports that instead of guessing.

3 Finding your way around

The rail on the left switches between five pages. Each page works the same way: choose a plugin, press the one large button, read the result.

PAGE	SHORTCUT	WHAT IT IS FOR
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Plugin Manager	⌘1 / Ctrl+1	Everything installed, by name, format and maker
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Graph Analyzer	⌘2 / Ctrl+2	Measure one plugin in detail
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Plugin Health

⌘3 / Ctrl+3

Thirteen robustness checks

A/B Compare

⌘4 / Ctrl+4

Two plugins, same measurements, side by side

Performance Profiler

⌘5 / Ctrl+5

CPU load, latency and memory

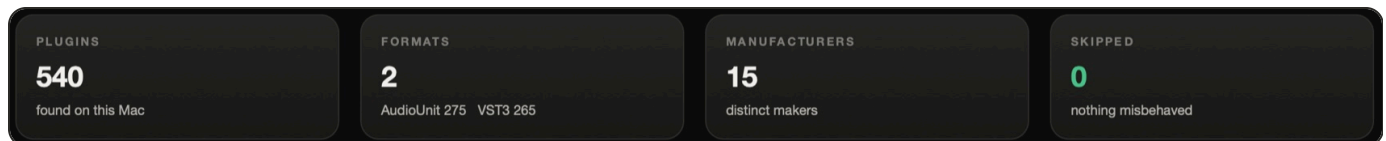
Choosing a plugin

Every page has the same picker. Type in the **Filter** box to narrow the list by name, maker or format, pick the plugin from the menu, then press **Load**. The line beside the button confirms what loaded: its channels and how many parameters it has.

A plugin with both an AudioUnit and a VST3 version appears twice. They are separate builds and can measure differently, so check you picked the one you use.

Around the edges

- **Sun or moon button** (top right): switches between the dark and light look. Your choice is remembered.
- **Status pill** (top right): what the library scan is doing.
- **Manual** (bottom of the rail): opens this document.
- **Update button** (bottom of the rail): appears only when a newer version exists. Click it to open the download page.
- **Plugin Manager → Reset skipped**: gives plugins that failed a scan another chance, for instance after you have activated them.



Plugin Manager's summary. "Skipped" counts plugins that crashed or hung while being scanned.

4 Graph Analyzer

Push test signals through a plugin and measure what comes back.



A distortion plugin, measured. Controls on the left, results on the right.

- 1 Pick a plugin and press **Load**.
- 2 Set the plugin how you want it measured. **Plugin UI** opens the plugin's own window; **Parameters** opens a plain list of every control with a filter, a reset and a randomise button.
- 3 Press **Run measurements**. It takes a second or two.
- 4 Read the five tiles, then the **Reading** card, then the graphs. Scroll for more graphs.

The measurement settings

CONTROL	WHAT IT DOES
Test tone	The frequency of the sine used for distortion, 1000 Hz unless you move it. Try 100 Hz on a bass processor or 5 kHz on a de-esser.
Buffer size	How many samples the plugin is handed at a time, like your DAW's buffer setting.
Sample rate	Measure at the rate you actually work at. Results can change with it.

Waveform shows	Which test signal the "Captured output" graph displays.
Re-run when settings change	Measures again whenever you move a control, so you can watch a curve respond while you turn a knob.
Also sweep distortion (slower)	Adds two graphs: distortion against level and distortion against frequency.

What it plays

Five signals, one after another: a single click (for frequency response, phase and the tail), a pure tone (for distortion and gain), two tones at once at 60 Hz and 7 kHz (for intermodulation), a level ramp (for dynamics), and silence (for self-noise). A sixth, a tone at 9 kHz when working at 48 kHz, is there to expose aliasing.

Instruments

A synth makes no sound from audio input, so VibeCheck plays it a MIDI note instead and measures the note. The frequency response of an instrument is not meaningful in the same way; look at the spectrum, the noise and the tail.

Keeping a result

- **Copy report** puts every number on the clipboard as text, ready to paste into a message or a note.
- **Save image** saves the results as a picture.

5 Reading the measurements

The ranges below are rules of thumb for judging a plugin that is meant to be clean. For a plugin that is meant to add character, high numbers are the point.

The five tiles

TILE	WHAT IT MEANS	HOW TO READ IT
THD	Total harmonic distortion: how much of the output is new overtones the input did not have. THD+N underneath includes noise too.	Under 0.01%: transparent. 0.01–0.1%: clean. 0.1–1%: audible colour. Over 1%: obvious saturation.
IMD	Intermodulation: new tones created when two notes play together. It is the harsher, less musical kind of distortion.	Read it like THD. A plugin with low THD but high IMD sounds clean on single notes and rough on chords.
Gain at 1 kHz	How much louder or quieter the plugin makes a mid-range tone.	0 dB is unity. Anything else matters when comparing: louder nearly always sounds "better".
Latency	The delay the plugin tells the host about, in milliseconds and samples.	0 is ideal for tracking. Larger values are normal for look-ahead limiters and linear-phase EQs; your DAW compensates.

Self-noise	What comes out when nothing goes in.	"Silent" or below -120 dBFS: inaudible. Above -90 dBFS: hiss you may hear when many instances stack up. Analog models often add noise on purpose.
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The Reading card

Each line is a plain-language summary of one family of measurements.

LINE	WHAT IT TELLS YOU	
Response	The overall tone: flat, or rolling off the lows or highs, and from where. "Flat within 0.1 dB from 40 Hz to 16 kHz" is a plugin that does not change tone at its current settings.	
Distortion	The character of the overtones. Even-order (H2, H4) reads as warm and tube-like. Odd-order (H3, H5) reads as harder and more aggressive. H2 and H3 are given in dBc: decibels below the tone itself.	
Dynamics	Whether gain changes with level. "Linear" means it does not. Otherwise you will see "Compressing from about -18 dB in, roughly 4:1", "Limiting", or "Soft saturation".	
Aliasing	False tones caused by distortion folding back from above the audible range. See the scale below.	
Tail	How long the plugin keeps sounding after the input stops. Milliseconds for an EQ, seconds for a reverb.	
Output level	Peak and average (RMS) level of the output, and the gap between them (crest).	

Aliasing scale

READING	LEVEL	MEANING
Clean	below -100 dBc	Nothing measurable that is not a harmonic.
Very low	-100 to -80	Inaudible in practice.
Low	-80 to -60	Faint, rarely audible.
Audible	-60 to -40	Can be heard on bright material such as cymbals and vocals.
Strong	above -40	The plugin aliases badly. Expect a gritty, digital edge when driven. Oversampling, if the plugin offers it, is the cure.

The graphs

Hover over any graph to read exact values. Frequency runs left to right on every graph that has one, from 20 Hz to 20 kHz.

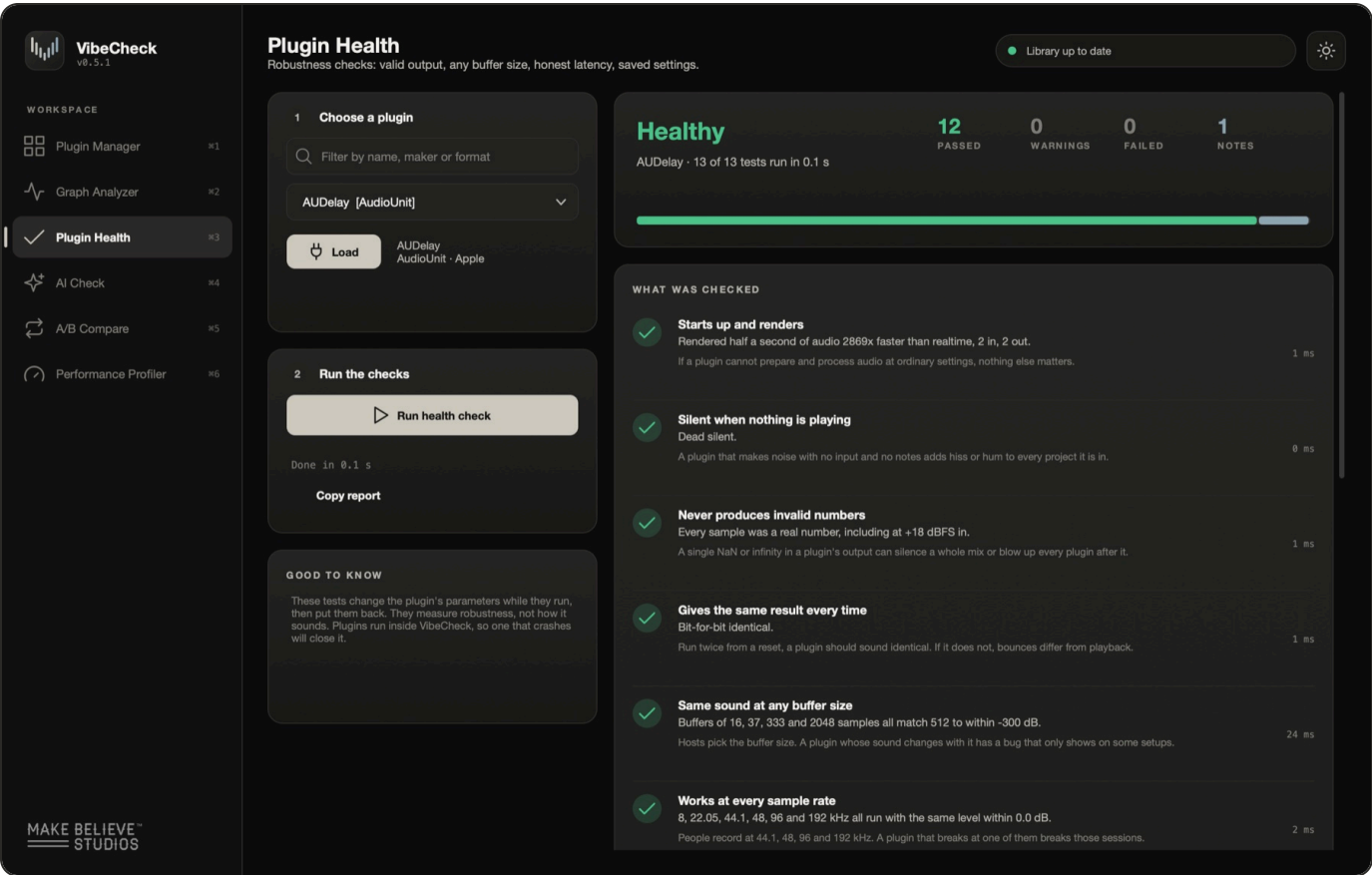
Magnitude response	Level against frequency: the EQ curve. A flat line means no tonal change. The menu smooths the curve (1/6 octave is a good default); choose less smoothing to see fine ripples.
Phase response	How much each frequency is delayed relative to the others. A straight or gently sloping line is benign. Sharp bends near a filter are normal for analog-style EQs and are part of their sound. A flat line with latency means linear phase.
Distortion spectrum	The output of the pure tone, as a spectrum. The tallest spike is the tone. Evenly spaced spikes to its right are harmonics. The dashed trace shows intermodulation products. Anything that is not on the regular grid is aliasing or noise.
Transfer curve	Level in against level out. A straight diagonal is linear. A line that bends over to the right is compression or limiting. Two traces, getting louder and getting quieter, that do not overlap show the plugin reacting to history, as compressors do.
Distortion against level	THD as the input gets louder. Flat and low: clean at any level. Rising to the right: it saturates when pushed, which is how to find the sweet spot of a saturator.
Distortion against frequency	THD across the spectrum. Rising toward the lows is common in transformer and tape models; rising toward the highs often points at aliasing.
Captured output	The actual waveform that came out, for the signal chosen under "Waveform shows".

MEASURE IT THE WAY YOU USE IT

Every number depends on the plugin's settings. A compressor at its default threshold may look perfectly linear because the test signal never reaches it. Set the plugin up as you would in a mix, then measure.

6 Plugin Health

Robustness checks: does the plugin behave, whatever the session throws at it?



Each check says what it found and, in grey, why it matters.

Load a plugin and press **Run health check**. The verdict at the top is one of:

VERDICT	MEANING
Healthy	Nothing failed, and at most one warning.
Mostly healthy	Nothing failed, but more than one thing deserves a look.
Has problems	At least one check failed in a way a user could hit.

The four counts beside it are **Passed**, **Warnings** (works, but look), **Failed** (broken) and **Notes** (facts, not judgements).

What is checked

CHECK	WHY A FAILURE MATTERS TO YOU
Starts up and renders	If it cannot process audio at ordinary settings, nothing else matters.

Silent when nothing is playing	Noise with no input adds hiss or hum to every project it is in.
Never produces invalid numbers	One bad value can silence a whole mix or blow up every plugin after it.
Gives the same result every time	If two runs differ, your bounce will not match what you heard.
Same sound at any buffer size	A plugin that changes with buffer size has a bug that shows only on some setups.
Works at every sample rate	From 8 to 192 kHz. Breaking at one rate breaks those sessions.
Reports its latency honestly	A wrong figure leaves that track audibly out of time.
No slowdown on near-silent input	Some plugins become many times slower in quiet passages and reverb tails.
Channel layouts	Mono tracks, stereo buses and surround each ask for a different layout.
Decays instead of ringing forever	A tail that never dies keeps the CPU busy and can build into feedback.
Parameters are well formed	Duplicate or empty names make automation and MIDI mapping confusing.
Saves and restores its settings	When a project reopens, the plugin must come back as you left it. This is the most common real-world bug.
Survives random settings	Eighteen extreme and random settings. It should stay sane with every control at once on minimum, maximum or anything between.

Instruments get one more: **Plays notes and lets go of them.**

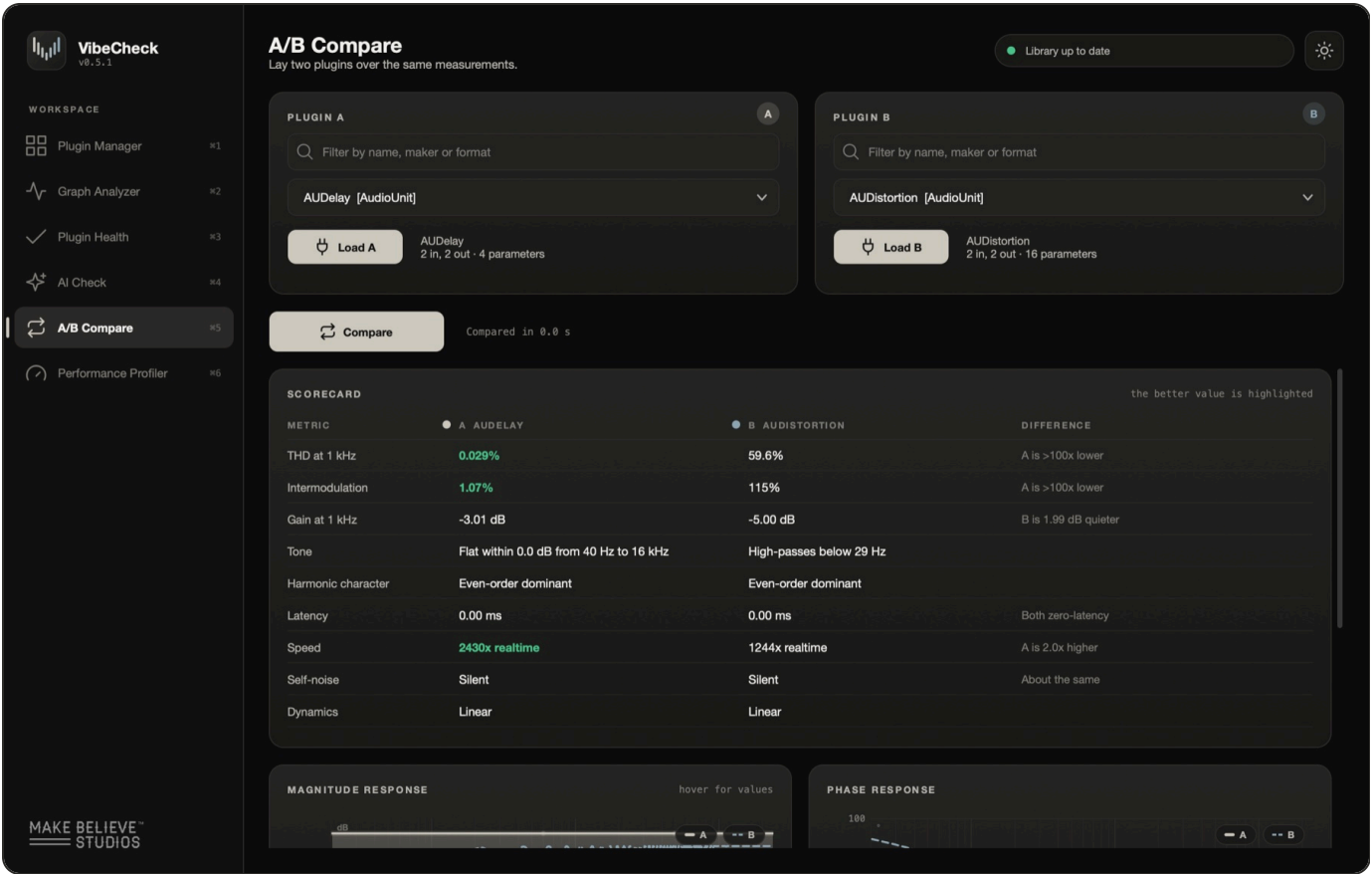
GOOD TO KNOW

These tests move the plugin's controls while they run, then put them back. They measure robustness, not sound. Plugins run inside VibeCheck here, so one that crashes outright will close the app; reopen it and the rest of your library is unaffected.

Copy report gives you the full list as text. If a plugin you paid for fails a check, that report is exactly what its developer needs.

7 A/B Compare

Lay two plugins over the same measurements and see precisely where they differ.



The scorecard. The better value in each row is shown in green, and the last column says by how much.

- 1 Choose a plugin under **Plugin A** and press **Load A**.
- 2 Choose another under **Plugin B** and press **Load B**.
- 3 Press **Compare**. Both get exactly the same test signals at the same settings.
- 4 Read the scorecard, then scroll to the four overlaid graphs.

The scorecard

ROW	WHAT IS COMPARED	BETTER IS
THD at 1 kHz	Harmonic distortion on a pure tone	Lower, if you want clean
Intermodulation	Distortion with two tones playing	Lower
Gain at 1 kHz	Level change through each plugin	Neither; it tells you to level-match
Tone	The shape of each frequency response, in words	Depends on what you want

Harmonic character	Even-order (warm) or odd-order (hard)	A matter of taste
Latency	Reported delay	Lower
Speed	How many times faster than real time each one runs	Higher means lighter on the CPU
Self-noise	Output with no input	Lower
Dynamics	Linear, compressing or limiting	Depends on the job

A value is only highlighted where one direction is clearly better. Tone and character are never highlighted, because neither is "right".

The overlaid graphs

Magnitude response, phase response, distortion spectrum and transfer curve, each with both plugins drawn together. **A is the solid line and B is the dashed line**, in the same colours as the dots in the scorecard header. Where the two lines sit on top of each other, the plugins do the same thing; where they part, they do not.

Making it a fair comparison

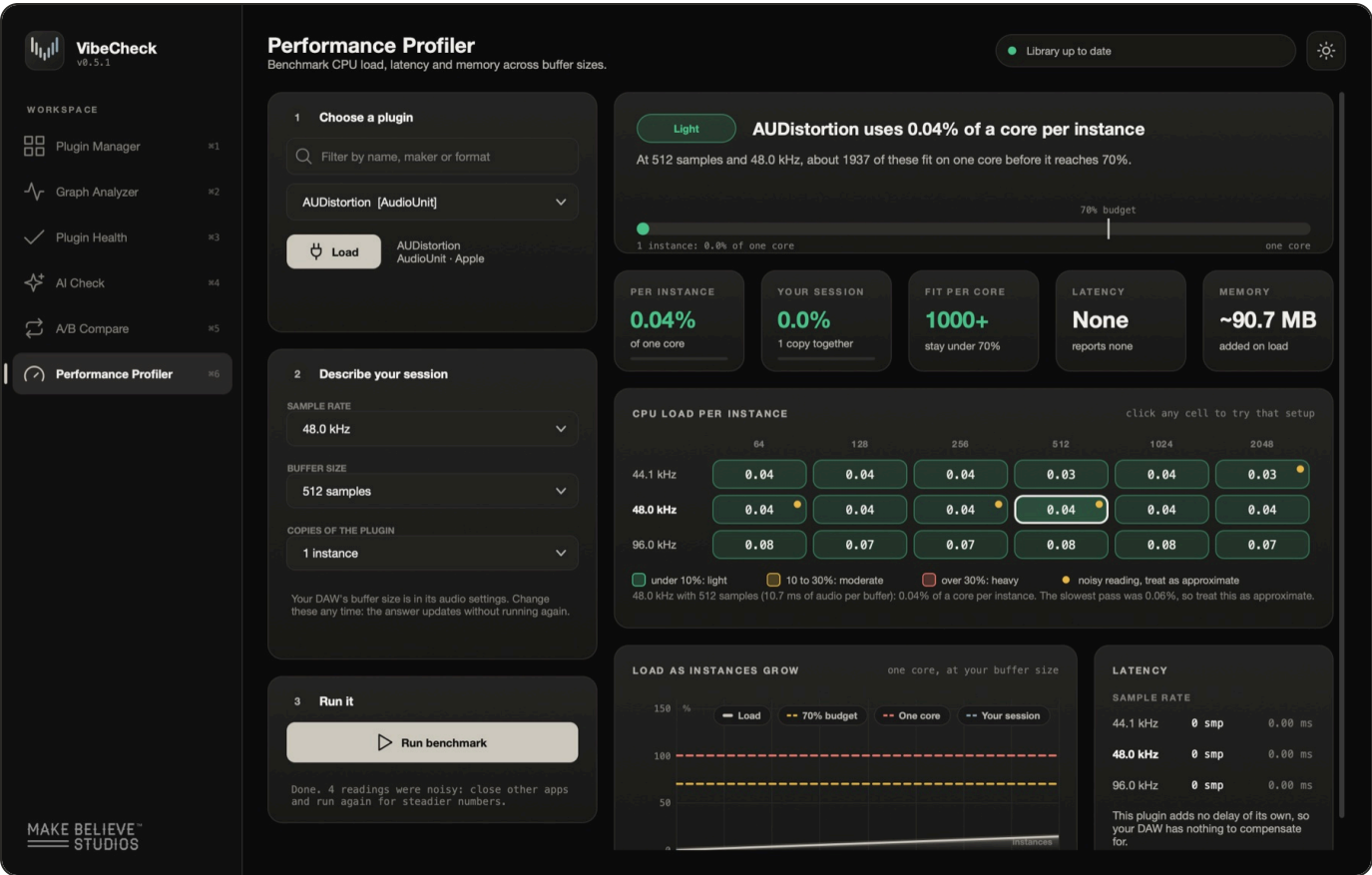
- **Match the settings first.** Two EQs at their defaults are both flat and will look identical. Set both to the same boost at the same frequency, then compare.
- **Mind the gain row.** If B is 2 dB quieter, it will sound duller in a listening test for that reason alone. Match the output levels before trusting your ears.
- **Compare like with like.** Use the same format for both where you can, and plugins that do the same job.
- **Small differences are noise.** THD of 0.002% against 0.003% is the same result. Look for differences of a factor of two or more, or a decibel or more in level.

DOES THE EXPENSIVE ONE DO ANYTHING THE FREE ONE DOES NOT?

Set both to the same curve and compare. If the magnitude and phase lines overlap and the distortion spectra match, they are the same processing, and you can choose on price, interface and CPU.

8 Performance Profiler

How much of your computer a plugin uses, and how many copies you can run.



The headline answers the question in one sentence. The grid shows every setup.

- 1 Choose a plugin and load it.
- 2 Describe your session: your sample rate, your DAW's buffer size, and how many copies you expect to use.
- 3 Run it with Run benchmark. It tests three sample rates and six buffer sizes.

After the first run you can change the session settings freely; the answer updates without measuring again.

Reading the result

PART	WHAT IT TELLS YOU
Headline	The share of one CPU core a single copy uses at your settings, rated Light (under 10%), Moderate (10–30%) or Heavy (over 30%).
Per instance	One copy's load, as a percentage of one core.
Your session	The load of all the copies you said you would use.

Fit per core	How many copies fit on one core while staying under 70%. The 70% line leaves headroom so a busy moment does not cause a dropout.
Latency	The delay the plugin reports. "None" means it adds no delay.
Memory	Roughly how much memory loading the plugin added.
CPU load per instance	The grid: one cell for each sample rate and buffer size. Greener is lighter. Click a cell to make that your session. A yellow dot marks a reading that varied between passes; treat it as approximate.
Load as instances grow	The line climbs as you add copies. Where it crosses the dashed 70% line is your practical limit; the 100% line is where one core is full.
Latency table	Reported delay at each sample rate, in samples and milliseconds.

ABOUT THE NUMBERS

These are measured on your computer, right now. Close other apps for steadier readings. Small buffers usually cost a little more per sample than large ones; a plugin that costs far more at 64 samples than at 1024 is doing heavy work on every buffer. Load roughly doubles from 48 to 96 kHz, because there is twice as much audio to process.

9 Recipes

Can I trust this plugin in a real session?

1. **Plugin Health** → Run health check. You want **Healthy**.
2. Any failure on *Saves and restores its settings*, *Same sound at any buffer size* or *Reports its latency honestly* is a reason not to use it on paid work.
3. **Performance Profiler** at your real buffer size, to check it fits.

Are these two EQs really different?

1. **A/B Compare**. Load both and set the same boost or cut on each.
2. Press Compare and look at the magnitude and phase graphs.
3. Lines that overlap mean the same processing. Differences in the phase graph, or in the distortion spectrum, are what "character" is made of.

Where is this saturator's sweet spot?

1. **Graph Analyzer**, with *Also sweep distortion* ticked.
2. Read *Distortion against level*: the level where the curve starts to climb is where it begins to work.
3. Check the *Distortion* line for even or odd character, and the *Aliasing* line. If aliasing reads Audible or Strong, try the plugin's oversampling setting and measure again.

Which of two compressors is cheaper to run?

1. **A/B Compare** for a quick answer: the *Speed* row.
2. **Performance Profiler** on each for the detail: *Fit per core* at your buffer size.

Why is my mix out of time after adding a plugin?

1. **Graph Analyzer**: compare the *Latency* tile with the "Delay" figure under *Gain at 1 kHz*. They should agree.
2. **Plugin Health**: *Reports its latency honestly* confirms it. If it fails, the plugin is delaying audio without telling your DAW.

10 Sharing results

One measurement

Copy report (Graph Analyzer, Plugin Health) and **Save image** (Graph Analyzer) are the quickest ways to send someone what you found.

Command line

For people who want to script it. On Windows use `vibecheck` if you ticked "Add to PATH" in the installer; on a Mac the installer package offers the same command.

```
vibecheck --health="Plugin Name"    run the health checks, print the result
vibecheck --analyze="Plugin Name"  measurements as text
vibecheck --selftest               check VibeCheck's own maths
```

A path to a plugin file works anywhere a name does, so a plugin need not be installed to be examined. `--health` exits with an error code when a check fails, which makes it usable in a build script.

11 Troubleshooting

WHAT YOU SEE	WHAT TO DO
A plugin is missing from the list	Check it is in a standard plugin folder, then relaunch so the library is checked again. On Windows only VST3 is supported.
A plugin is listed as skipped	It crashed or hung while being scanned, often because it wanted a licence. Activate it, then use Plugin Manager → Reset skipped .
A plugin loads but every measurement is silence	It is probably waiting for a licence or for hardware and is passing no audio. Open Plugin UI to see whether it is asking for something.
A compressor measures as "Linear"	The test signal is not reaching its threshold. Lower the threshold, or raise the input gain, and measure again.
Performance numbers change between runs	Other programs are competing for the CPU. Close them and run again. Cells with a yellow dot were noisy.
The Mac says the app cannot be opened	Right-click the app and choose Open, or use System Settings → Privacy & Security → Open Anyway.
VibeCheck closed during a health check	The plugin crashed and took the app with it. That is itself the finding. Reopen VibeCheck.

Uninstalling VibeCheck

- **Mac:** open the disk image and double-click **Uninstall VibeCheck**, or drag the app to the Trash.
- **Windows:** Settings → Apps → Installed apps → VibeCheck → Uninstall.

12 Glossary

TERM	MEANING
Aliasing	False tones that appear when distortion creates frequencies too high for the sample rate, which then fold back into the audible range.

Buffer size	How many samples the computer processes in one go. Smaller means less delay when playing, and more CPU.
dB	Decibel: a unit of relative level. +6 dB is roughly twice the amplitude.
dBFS	Level relative to the loudest a digital signal can be. 0 dBFS is the ceiling; everything else is negative.
dBc	Level relative to the test tone ("the carrier"). -60 dBc is one thousandth of the tone's amplitude.
Denormals	Extremely small numbers that some processors handle very slowly, causing CPU spikes in near-silence.
Harmonics	Overtones at whole multiples of a note's frequency. H2 is twice the frequency, H3 three times.
IMD	Intermodulation distortion: tones created by two notes interacting, at frequencies that belong to neither.
Instance	One copy of a plugin in a session.
Latency	Delay added by a plugin. Hosts compensate for the amount the plugin reports.
Linear phase	Every frequency delayed by the same amount, at the cost of latency.
RMS	A measure of average level, closer to perceived loudness than the peak.
Sample rate	How many times per second audio is measured: 44.1, 48, 96 kHz and so on.
THD	Total harmonic distortion: the combined level of all the harmonics a plugin adds, as a percentage of the tone.
Unity gain	Output level equal to input level: 0 dB of change.

VibeCheck is made by Make Believe Studios. It checks once at launch whether a newer version exists by reading one small public file, and sends nothing about you or your plugins.